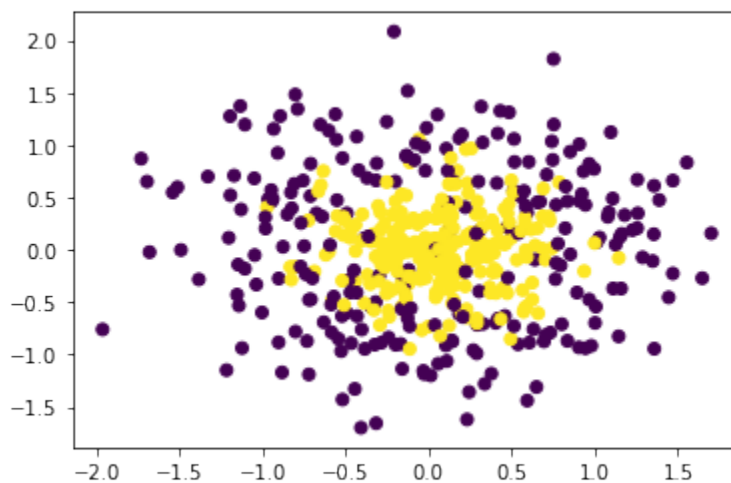


```
In [87]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from mlxtend.plotting import plot_decision_regions
from sklearn.model_selection import train_test_split
from sklearn.datasets import make_circles
```

```
In [88]: np.random.seed(42)
X, y = make_circles(n_samples=500, factor=0.1, noise=0.35, random_state=42)
#X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)
```

```
In [89]: plt.scatter(X[:,0],X[:,1],c=y)
```

Out[89]: <matplotlib.collections.PathCollection at 0x2b28c5fa700>



```
In [90]: from sklearn.ensemble import AdaBoostClassifier
from sklearn.model_selection import cross_val_score

abc = AdaBoostClassifier()

np.mean(cross_val_score(abc,X,y,scoring='accuracy',cv=10))
```

Out[90]: 0.786

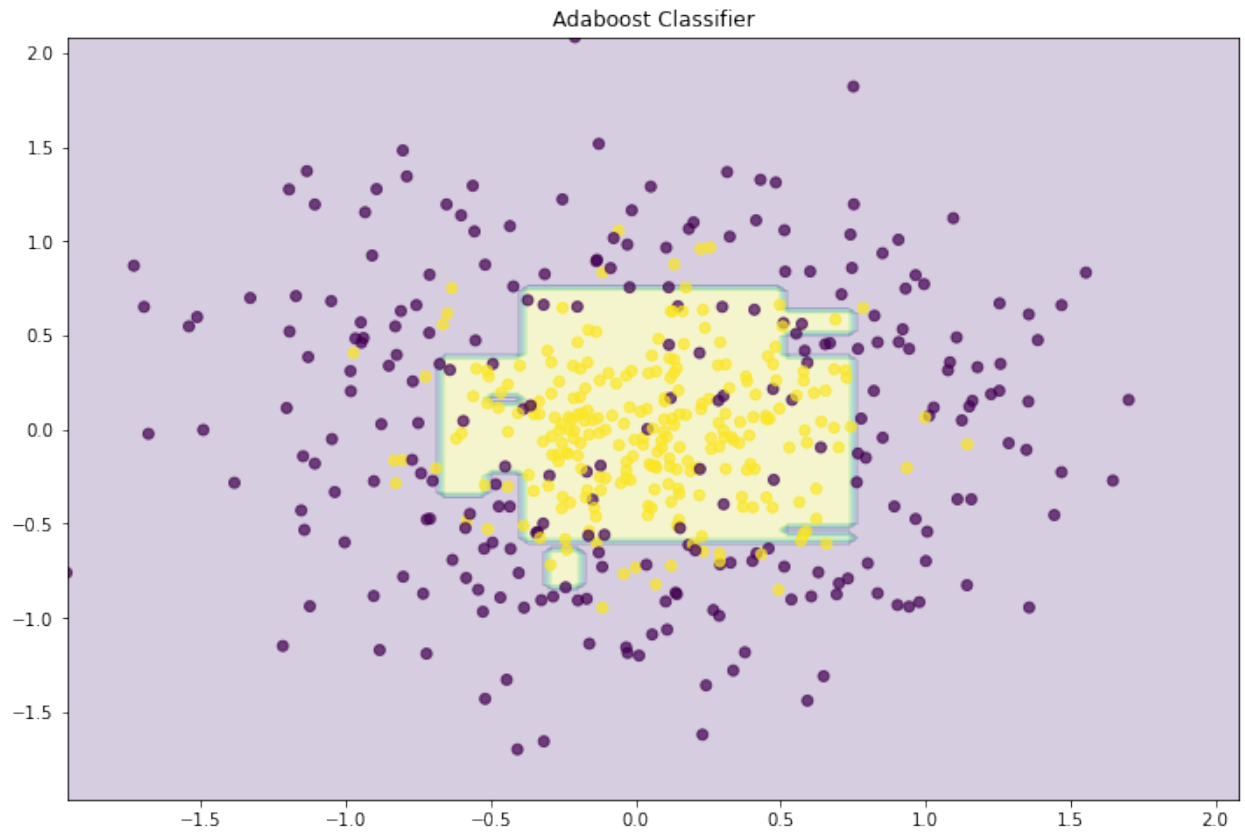
```
In [91]: abc.fit(X,y)
```

Out[91]: AdaBoostClassifier()

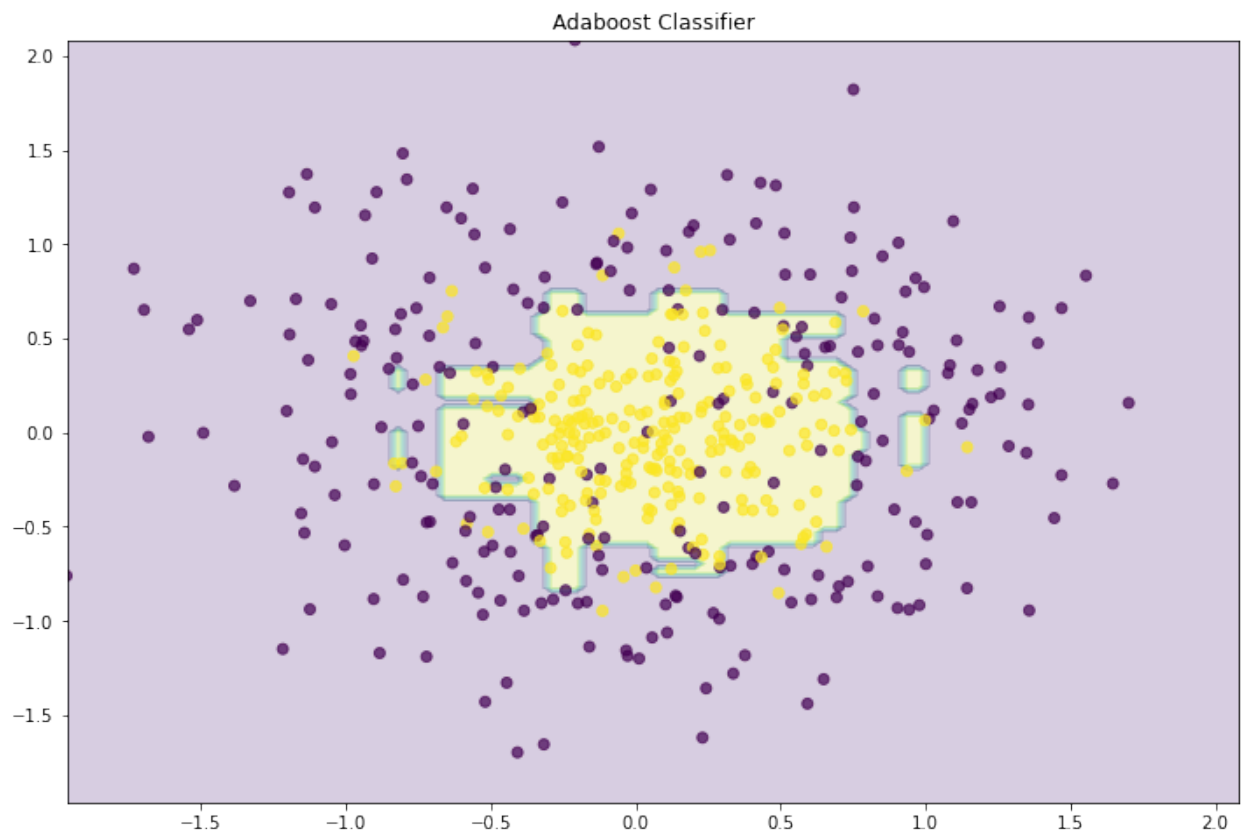
```
In [92]: def plot_decision_boundary(clf):
plt.figure(figsize=(12, 8))
x_range = np.linspace(X.min(), X.max(), 100)
xx1, xx2 = np.meshgrid(x_range, x_range)
y_hat = clf.predict(np.c_[xx1.ravel(), xx2.ravel()])
y_hat = y_hat.reshape(xx1.shape)
plt.contourf(xx1, xx2, y_hat, alpha=0.2)
plt.scatter(X[:,0], X[:,1], c=y, cmap='viridis', alpha=.7)
```

```
plt.title("Adaboost Classifier")
plt.show()

plot_decision_boundary(abc)
```



```
In [99]: abc = AdaBoostClassifier(n_estimators=1500, learning_rate=0.1)
abc.fit(X,y)
plot_decision_boundary(abc)
```



In []:

In [76]: **from** sklearn.model_selection **import** GridSearchCV

```
grid = dict()
grid['n_estimators'] = [10, 50, 100, 500]
grid['learning_rate'] = [0.0001, 0.001, 0.01, 0.1, 1.0]
grid['algorithm'] = ['SAMME', 'SAMME.R']
```

```
grid_search = GridSearchCV(estimator=AdaBoostClassifier(), param_grid=grid, n_
# execute the grid search
grid_result = grid_search.fit(X, y)
# summarize the best score and configuration
print("Best: %f using %s" % (grid_result.best_score_, grid_result.best_params_
```

Best: 0.832000 using {'algorithm': 'SAMME', 'learning_rate': 0.1, 'n_estimators': 500}

In []: # Link
<https://stats.stackexchange.com/questions/82323/shrinkage-parameter-in-adaboos>